CAD ENGINEERING TECHNICAL INTERVIEW

EDA / CAD Engineering Projects

Logic Synthesis / Physical Design / Timing ECO / CAD Flow Automation

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Education & Research Interests

01 / PROFILE

Education

National Taiwan University
M.S. Candidate, Data Science Master Degree Program
Sep. 2025 – Expected Jun. 2027

National Chengchi University
B.S. in Computer Science
Double Major in Electronic Physics
Sep. 2021 – Jun. 2025

Research Interests

EDA / CAD & VLSI Design Automation

Logic Synthesis
Technology Mapping
Static Timing Analysis
Timing Closure
Physical Design
Incremental ECO
PPA

Algorithm & Software Development

C++ / Python / Tcl Flow Automation

Professional Software Engineering Experience

02 / WORK EXPERIENCE

Software development, simulation, and validation experience in R&D environments.

AIMobile Co., Ltd. (Inventec Group)

Apr. 2025 – Jul. 2025

Contract Software Engineer

Contributed to software development and system integration in an R&D environment.

Supported debugging and functional validation across software modules.

Neopower Technologies Co., Ltd.

Jun. 2023 – Mar. 2025

Contract Software Engineer

Developed a custom system simulator to reproduce operational and error scenarios.

Performed regression checks, log analysis, and customer-side A/B validation.

Selected Projects & Discussion Topics

03 / PROJECT OVERVIEW

01 Master's Research — Logic Synthesis (Front End)

Joint Single- and Multi-Output Cell Mapping via Differentiable Optimization

EDA Tool Development / Boolean Matching / Area–Delay Optimization

02 ICCAD 2025 Problem C — CAD Methodology (Back End)

STA-Driven Incremental Timing ECO Search

OpenROAD / OpenSTA / Python–Tcl Flow Automation / Heuristic PPA Search

Joint SO / MO Mapping: Research Pipeline

01 / RESEARCH SCOPE

Discover FA/HA opportunities, pre-select non-overlapping MO candidates, then let the optimizer choose SO or BIND.

Problem

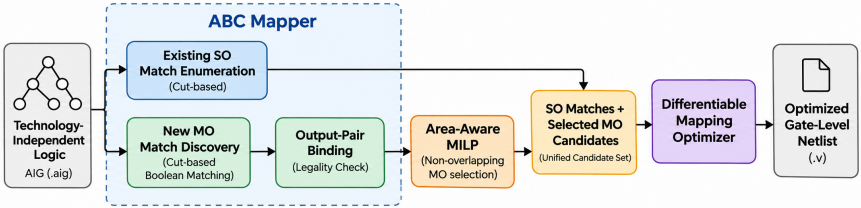
Single-output cut retention does not target two-output FA/HA opportunities.

ABC-Based MO Discovery

A new MO candidate search inside the ABC Mapper. Existing SO enumeration stays unchanged.

MILP Pre-selection

Keep a non-overlapping, area-aware subset of legal MO candidates. SO matches bypass this step.



Why an Independent MO Discovery Path?

03 / MOTIVATION

SO-oriented cut pruning may remove cuts useful for joint FA/HA output matching.

ABC Mapper SO Ranking

Cuts are retained for single-output area and delay, not for complementary FA/HA output pairs.

ABC-Based MO Discovery

Keep additional MO cuts and CON/SN truth projections without changing the ABC Mapper's SO path.

FIGURE SPACE

SO-pruned cut vs MO-retained cut

Two output roots with a shared legal 3-leaf input support.

MO discovery is not equivalent to scanning the final selected SO cover.

MO Candidate Discovery in ABC

04 / CORE ALGORITHM

Pair compatible CON and SN projections over a common input support.

01 / MO Cut Propagation

Independent retention of FA/HA-relevant truth functions and supports.

02 / Projection Pairing

Match CON / SN roles using shared leaves, permutation and phase.

03 / Legality + Witness

Validate the pair and record the SO cells this MO candidate would replace.

FIGURE SPACE

Example: FA CON / SN cut projections

Highlight two output roots, same leaves, shared cell configuration.

OUTPUT: BIND / ROOTS / FANINS / REPLACE_ROOTS

Baseline: File-Order Candidate Loss

File-order greedy filtering ignores estimated area benefits of conflicting MO candidates.

41,476
Original MO candidate BIND records

7,024
FIFO retained (offline simulation)

16.9%
Overall retention rate

Design	Original MO	FIFO Retained	Dropped	Retention
EPFL hyp	25,440	4,180	21,260	16.4%
EPFL square	3,658	544	3,114	14.9%
EPFL log2	2,798	400	2,398	14.3%
EPFL multiplier	2,190	349	1,841	15.9%
EPFL voter	632	106	526	16.8%
ISCAS c6288	258	37	221	14.3%
ISCAS c7552	102	11	91	10.8%
EPFL adder	6	1	5	16.7%
All 364 designs	**41,476**	**7,024**	**34,452**	**16.9%**

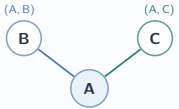
Source: MO-candidate census, 364 benchmark entries; simulated first-fit retention, not mapping-optimizer QoR.

Area-Aware MO Selection (MILP)

09 / GLOBAL CANDIDATE SELECTION

Maximize estimated area savings subject to non-overlapping MO candidates.

$$
\begin{aligned}
\max \quad & \sum_{i=1}^{N} \left(A_i^{SO} - A_i^{MO}\right) x_i \\
\text{s.t.} \quad & \sum_{i=1}^{N} a_{vi} x_i \le 1, \quad \forall v \in \mathcal{V} \\
& x_i \in \{0,1\}, \quad i = 1, \ldots, N
\end{aligned}
$$



$x_{AB} + x_{AC} \le 1$

Notation

x_i: select MO i; A_i^{SO}: SO witness area; A_i^{MO}: MO cell area;

$a_{vi} = 1$ iff AIG base node v belongs to MO candidate i; otherwise 0.

Implementation

Python + HiGHS (`highspy`); emit selected.matches.

Objective weights estimate local SO-to-MO benefits. Final QoR is measured after the mapping optimizer; MILP retention does not force every selected candidate into the netlist.

Current Mapping Flow & ADP Improvement

11 / CURRENT RESULTS

Current flow

MO candidate discovery in ABC

Boolean / structural legality

SO witnesses + area weights

MILP candidate pre-selection

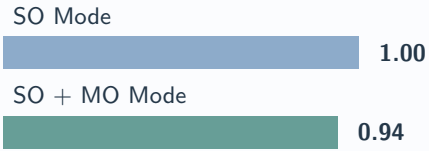
Mapping optimizer SO / BIND choice

Measure final mapped QoR after optimization; MILP weights are local estimates.

6%

Lower ADP vs. SO Mode

Normalized ADP (SO Mode = 1.00)



ADP = area × delay. Comparison is SO mode versus SO+MO mode.

PROJECT 02 / ICCAD 2025 CAD CONTEST / PROBLEM C

Incremental Placement Optimization Beyond Detailed Placement

Simultaneous Gate Sizing, Buffering, and Cell Relocation

MY PROPOSED APPROACH

STA-Driven Incremental Timing ECO Search

Use OpenROAD / OpenSTA with policy-controlled ECO moves and global PPA evaluation.

Multi-Objective Timing ECO Optimization

02 / PROBLEM DEFINITION

Improve setup TNS toward zero while controlling total power, HPWL, and cell displacement.

OPTIMIZATION OBJECTIVES

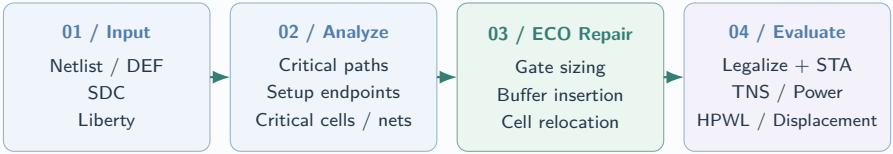
01 / Setup TNS
Reduce negative setup slack at violating endpoints.

02 / Total Power
Avoid excess power from ECO changes.

03 / HPWL
Control estimated wirelength after ECO.

04 / Displacement
Limit cell movement and keep placement legal.

BASIC ECO EVALUATION PIPELINE



Key challenge: a timing fix can worsen power or physical cost.

Five Automatic ECO Strategy Candidates

02 / CANDIDATE SET

The candidate is an automatic repair policy and editable region, not a manually chosen cell replacement.

S1 / Global Setup

Whole design; automatic upsizing and VT selection.

S2 / Path-Restricted

Top-K path cells + nets; localized setup repair.

S3 / Overlap-Aware

Top-I shared cells, nets fixed; sizing / VT repair.

S4 / Local Buffering

Top-K nets, original cells fixed; buffer insertion and load splitting.

S5 / Power Recovery

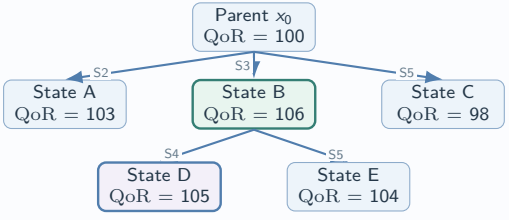
Slack-rich cells / wider region; lower-power equivalent cells.

STA-Guided ECO Search: Best-First Branch List

02 / HEURISTIC SEARCH POLICY

No better child this round? Continue from the best unexplored state in the Branch List.

Search Tree Example



Bounded Branch List: D:105 > E:104 > A:103

Global Best: B (106) | **Next:** D (105)

A lower-scoring intermediate state may lead to a better ECO sequence later.

Heuristic Search Steps

01 / Pop Parent

Highest-QoR unexplored state from the **Branch List**.

02 / Generate + Evaluate

STA-guided ECO moves, legalization, and global QoR score.

03 / Update States

Save improved **Global Best**; queue legal child states.

04 / Next Branch

Expand best branch, **even with no new Global Best**.

QoR Results & Conclusion

02 / EARLIER ECO EVALUATION

Setup-focused timing repair and the resulting power / wirelength trade-off.

Setup Timing

16.7%
Median Setup TNS Improvement

14.6%
Average Setup WNS Improvement

Power & Wirelength

8%
Total Power Reduction

+2%
HPWL Increase

Conclusion

ECO operators can improve setup timing and power while accepting a small wirelength trade-off.

THANK YOU

Questions & Discussion